

# Yihan Li

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## Education

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| <b>Sun Yat-sen University</b>                                                 | Sep 2023 – Jul 2027 (Expected) |
| <i>B.A. in English Language &amp; Literature, School of Foreign Languages</i> | <i>Major GPA: 3.8/4.0</i>      |
| <i>Minor in Intelligent Unmanned Systems</i>                                  | <i>GPA: 4.0/4.0</i>            |
| <i>Minor in Computer Science &amp; Technology</i>                             | <i>GPA: 3.8/4.0</i>            |

## Publications

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|-------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>Anticipating Contact for Visuo-Tactile Diffusion Policies</b>                                                              | Under Review |
| Yating Feng, Yuhang Zheng, Lixin Xu, <b>Yihan Li</b> , Jiaju Yin, Renjing Xu                                                  |              |
| <b>Goldsmith: Gold-Loss-Guided Prompt Training for Agentic Annotation</b>                                                     | Under Review |
| <b>Yihan Li</b> , Hanyi Zhang, Xiaoxi Jiang, Man Guo                                                                          |              |
| <b>Unordered Landmark Visual Navigation</b>                                                                                   | ECCV 2026    |
| Hao Ren, Junzhe Zhu, <b>Yihan Li</b> , Zetong Bi, Le Zheng, Zhi Li, Yiqing Yuan, Zhaoliang Wan, Lu Qi, Hui Cheng              |              |
| <b>OmniTrav: A Dataset and Benchmark for 360° Vision-based Traversability via Foundation Models</b>                           | IROS 2026    |
| <b>Yihan Li</b> , Hao Ren, Zhenglan Jiang, Junzhe Zhu, Hui Cheng                                                              |              |
| <b>RAPID Hand: A Robust, Affordable, Perception-Integrated, Dexterous Manipulation Platform for Generalist Robot Autonomy</b> | NeurIPS 2025 |
| Zhaoliang Wan, Zetong Bi, Zida Zhou, Hao Ren, Yiming Zeng, <b>Yihan Li</b> , Lu Qi, Xu Yang, Ming-Hsuan Yang, Hui Cheng       |              |

## Work Experience

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| <b>HCL Lab (HKUST(GZ))</b>                                                                                                                                                                                                                                                                                                                          | Jan 2026 – Present      |
| <i>Research Assistant</i>                                                                                                                                                                                                                                                                                                                           | <i>Prof. Renjing Xu</i> |
| <ul style="list-style-type: none"><li>Researching tactile sensing, vision models, VLA fine-tuning, VLA-based online RL, and Real2Sim scene reproduction for dexterous manipulation.</li><li>Exploring multimodal visuo-tactile policy learning and evaluation protocols for embodied manipulation under real-world sensing constraints.</li></ul>   |                         |
| <b>Yun Drone</b>                                                                                                                                                                                                                                                                                                                                    | Feb 2025 – Present      |
| <i>UAV Development Intern</i>                                                                                                                                                                                                                                                                                                                       |                         |
| <ul style="list-style-type: none"><li>Worked on UAV autonomy development across propulsion selection, formation flight, navigation and localization, and flight-test workflow design.</li><li>Studied RL-based disturbance-resilient control and supported system iteration through flight-log analysis and control tuning.</li></ul>               |                         |
| <b>RAPID Lab (SYSU)</b>                                                                                                                                                                                                                                                                                                                             | Nov 2024 – Mar 2026     |
| <i>Research Assistant</i>                                                                                                                                                                                                                                                                                                                           | <i>Prof. Hui Cheng</i>  |
| <ul style="list-style-type: none"><li>Conducted embodied intelligence research across dexterous manipulation, multimodal perception, and vision-based navigation/traversability.</li><li>Supported RAPID Hand, OmniTrav, and ULVN research directions through experimental validation, perception integration, and benchmark development.</li></ul> |                         |
| <b>RoboTech Association (SYSU)</b>                                                                                                                                                                                                                                                                                                                  | Sep 2024 – Dec 2025     |
| <i>President; Competition Team Captain</i>                                                                                                                                                                                                                                                                                                          |                         |
| <ul style="list-style-type: none"><li>Led robotics training and competition preparation across mechanical design, hardware, embedded control, vision, navigation, and algorithms.</li><li>Coordinated association operations, technical mentoring, cross-group collaboration, and onboarding materials for 160+ members.</li></ul>                  |                         |

## Projects

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### **Venom VNV: ROS 2 Unified Robotics Platform**

Nov 2025 – Present

*Project Lead; ROS 2 Platform Developer*

[venom-algorithm.github.io/Venom\\_VNV](https://venom-algorithm.github.io/Venom_VNV)

- Architected a ROS 2 Humble platform spanning drivers, perception, localization, planning, mission control, simulation, and interface specifications for UGV/UAV/USV scenarios.
- Integrated Livox MID360, Hikvision cameras, AgileX chassis/arm/PX4 drivers with Point-LIO/Fast-LIO, Nav2 TEB, Ego Planner Swarm, YOLO detection, auto-aim tracking, and waypoint task management.

### **RAPID Hand: Perception-Integrated Dexterous Manipulation Platform**

Nov 2024 – Jun 2025

*Research Assistant*

[rapid-hand.github.io](https://rapid-hand.github.io)

- Co-developed a compact 20-DoF dexterous hand ontology and high-DoF teleoperation/data-collection stack for contact-rich whole-hand manipulation.
- Integrated hardware-level whole-hand perception, synchronizing wrist vision, fingertip tactile sensing, and proprioception with sub-7 ms latency for policy learning.

### **OmniTrav: 360° Vision-based Traversability Benchmark**

Jun 2025 – Mar 2026

*First Author*

- Proposed the C2T pipeline and built OmniTrav, a 4.7k-scene, 250k-frame, 577 GB dataset mapping pinhole, fisheye, and panoramic RGB to planner-ready 360-degree radial clearance.
- Designed an automated foundation-model annotation pipeline and a DINOv3-FPN baseline with a cross-attention angle projector for radial distance and obstacle prediction.

### **GenreShift and Rosetta: LLM Systems for Linguistic Research**

Oct 2024 – Present

*Project Leader*

[genreshift.cn](https://genreshift.cn)

[rosetta-stone.xyz](https://rosetta-stone.xyz)

- Built GenreShift for genre and discipline analysis, extracting quantitative linguistic features such as TTR, MTLT, academic vocabulary, and cross-genre comparisons for LoRA/MoE-based genre transformation; filed invention patent application No. 2025105553019.
- Developed Rosetta as an agentic annotation pipeline with concept bootstrap, prompt optimization, embedding-based top-k retrieval, LLM judge routing, review feedback, and Prodigy-compatible JSONL export.

## Awards

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### **China Robot and Artificial Intelligence Competition**

2025

National First Prize (Team Captain)

### **National Intelligent Unmanned Systems Application Challenge**

2025

National Champion (Team Captain)

### **National Undergraduate Electronics Design Contest**

2025

National Second Prize (Team Captain)

### **National Undergraduate Robotics Competition - RoboMaster University League**

2025

National Second Prize (Vision Group Leader)

## Skills

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|---------------------------|----------------------------------------------------------------------|
| <b>Programming:</b>       | Python, C/C++, MATLAB, STM32/Arduino.                                |
| <b>Robotics and Sim:</b>  | ROS1/2, MuJoCo, Isaac Sim, Isaac Lab, Gazebo, PX4, Hermes, OpenClaw. |
| <b>Development Tools:</b> | Codex, Claude Code.                                                  |
| <b>Engineering Tools:</b> | Fusion 360, SolidWorks, EDA design; AOPA Multi-rotor VLOS Pilot.     |
| <b>Languages:</b>         | Mandarin (native), English (TEM-4), French.                          |